

Lab 05: NAT Configuration

Duration: 1 hours 30 minutes | In-Lab Assessment

Tools: Cisco Packet Tracer

Objectives

By the end of this lab, students will be able to:

- Configure Static, Dynamic, and Port-Address Translation (PAT) NAT on Cisco routers.
- Verify NAT translations and statistics.
- Learn about access control list (ACL)

Lab Description

Part A: Network Address Translation (NAT)

NAT allows internal hosts to use private IPs locally while sharing a few public IPs for Internet access. The gateway router translates between private and public IPs, hiding internal addresses from the outside world. This conserves global IPs and enables many devices to connect using limited public addresses.

There are three types of NAT that you can define in Cisco devices: Static, Dynamic, and Overloaded or Port Address Translation (PAT).

Part B: Static NAT

It allows one-to-one mapping between local and global addresses. You will have to configure one global IP address for each internal host you want NAT to translate. The command to enable the static translation is as follows:

```
Router(config)# ip nat inside source static local_ip global_ip
```

You must designate which router ports face your private LAN (ip nat inside) and which face the public network (ip nat outside) so the router knows where to apply source-based (inside→outside) and destination-based (outside→inside) translations. You can tag multiple interfaces as inside or outside, and NAT begins translating once at least one of each is configured—this applies to all NAT types. Following are the commands to specify the inside and outside interfaces:

```
1 Router(config-if)# ip nat inside
2 Router(config-if)# ip nat outside
```

How translation actually works:

1. Outbound (LAN → Internet)

- Packet arrives on an ip nat inside interface with source 192.168.1.25.
- Router looks at your NAT rules and replaces that source with a public IP (e.g. 203.0.113.10 or a port-mapped version of it).

- Packet then egresses on an ip nat outside interface toward the Internet.

2. Inbound (Internet → LAN)

- Response packet arrives on the ip nat outside interface addressed to your public IP (or to one of your pool addresses / port).
- Router looks up the translation table and rewrites the destination back to the original private IP (e.g. 192.168.1.25), then forwards it out the ip nat inside interface

Part C: Access Control List (ACL)

A Cisco IP ACL (Access Control List) is simply a numbered (or named) list of “permit”/“deny” statements that match packets by IP fields. Standard ACLs (numbers 1–99, 1300–1999) inspect only the source address; extended ACLs (100–199, 2000–2699) can match source, destination, protocol, ports, etc.

There are two steps to implement an ACL. First, define the rule. Second, apply the rule to an interface.

The command format for defining a numbered standard IP ACL is:

```
1 Router(config)# access-list access_list_number {permit|deny}
2 {(source_address source_wildcard)|any}
```

Remember that whenever you apply an ACL to an interface, all the traffic that does not match any ACL rule will be discarded by default.

For example, you have defined an ACL to deny a certain source IP. Whenever you apply that rule to an interface, all packets other than the denied source will also be discarded because there is no matching rule for those packets.

So, you must allow other traffic explicitly by defining another ACL. The *any* keyword is handy in this case. To permit (or deny) any packet other than the previously specified rules, you can add the keyword *any* in place of the source_address and source_wildcard like the following:

```
1 Router(config)# access-list 1 permit any
```

Another thing is you can only use numbers from 1 to 99 to specify the access list number. Other numbers are used for extended numbered ACLs. After defining the ACL rule, we must apply it to an interface. Remember that the ACL has no effect until you apply it. The command format for applying an ACL to an interface is:

```
1 Router(config-if)# ip access-group access_list_number {in|out}
```

The ACL is applied either for inbound or outbound traffic of an interface, and you need to specify the corresponding keyword, i.e., *in* or *out* for that. One best practice before applying an ACL to an interface is to verify the rule by using the following command:

```
1 Router# show access-lists
```

Part D: Dynamic NAT

Dynamic NAT automatically maps each internal host to a temporary public IP from a pre-defined pool, releasing the address after a timeout so others can use it. It's simpler than static NAT because you just define an access list of local addresses and a global IP pool. The command format for defining a numbered standard IP ACL is:

```
1 Router(config)# access-list access_list_number permit source_address
2                      source_wildcard
```

Then, you have to specify the pool of global IP addresses from where the IPs will be allocated. The pool is a range of IP addresses in a given network where the subnet mask will specify the corresponding network portion. The command to specify the pool is as follows:

```
1 Router(config)# ip nat pool POOL_NAME start_ip end_ip netmask
   subnet_mask
```

Then, you must establish the relation between the earlier defined access list and the nat pool through the following command:

```
1 Router(config)# ip nat inside source list access_list_number pool
   POOL_NAME
```

After that, you must specify the inside and outside interfaces, such as the static NAT.

Part E: Port Address Translation (PAT)

PAT (Port Address Translation) lets hundreds of internal hosts share one public IP by tagging each connection with a unique TCP/UDP port. Configure it like dynamic NAT but add the overload keyword to your ip nat inside source command to enable many-to-one mapping. The command format for the configuration of PAT is as follows:

```
1 Router(config)# ip nat inside source list access_list_number pool
   POOL_NAME
2                      overload
```

You can also configure PAT using the IP address of an interface rather than a NAT pool. In this case, the local IPs will be translated to the interface's IP address.

```
1 Router(config)# ip nat inside source list access_list_number interface
2                      Interface_NAME_Connected_to_Outside_Networks overload
```

Part F: Configure ACL

A router of device model 2911, three switches of device model 2960, and three PCs have been used in the sample topology shown in Figure 2 for the ACL configuration.

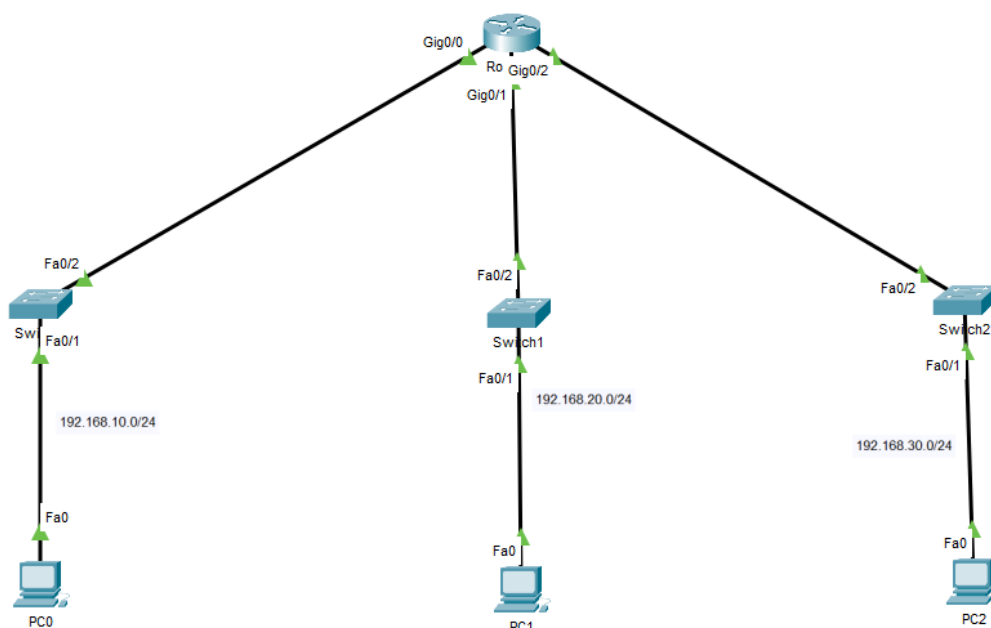


Figure 2: A sample network topology for the configuration of ACL.

1. Configure R1 Interfaces

```

1 Router(config)# int g0/0
2 Router(config-if)# ip address 192.168.10.1 255.255.255.0
3 Router(config-if)# no shutdown
4 Router(config-if)# exit
5 Router(config)# int g0/1
6 Router(config-if)# ip address 192.168.20.1 255.255.255.0
7 Router(config-if)# no shutdown
8 Router(config-if)# exit
9 Router(config)# int g0/2
10 Router(config-if)# ip address 192.168.30.1 255.255.255.0
11 Router(config-if)# no shutdown
12
13 Router(config-if)# end
14 Router# copy running-config startup-config

```

2. Configure PC0

```

1 IP: 192.168.10.5
2 Mask: 255.255.255.0
3 Gateway: 192.168.10.1

```

3. Configure PC1

```

1 IP: 192.168.20.5
2 Mask: 255.255.255.0
3 Gateway: 192.168.20.1

```

4. Configure PC1

```
1 IP: 192.168.30.5
2 Mask: 255.255.255.0
3 Gateway: 192.168.30.1
```

5. Define ACL

```
1 Router(config)# access-list 1 deny 192.168.10.0 0.0.0.255
2 Router(config)# access-list 1 permit any
```

6. Verify ACL

```
1 Router# show access-lists
```

7. Apply ACL

```
1 Router(config)# interface g0/2
2 Router(config-if)# ip access-group 1 out
```

8. Verify if ACL has been applied to the router interface

- (a) Go to the simulation mode.
- (b) From the event list, allow only ICMP-type messages.
- (c) Ping a message from PC0 to PC2.
- (d) The message will not be sent to the PC1 from the router. Instead, it will send back a notification message to PC0.

Part G: Configure Static NAT

Two routers of device model 2811, one switch of device model 2960, two PCs, and one server have been used in the sample topology shown in [Figure 3](#) for the NAT configuration.

The objective is, Server0 has private address 10.0.0.5 which needs to be translated to 50.0.0.5. In the Router 1 we need to apply NAT configuration to achieve it.

From PC0 or PC1 we can send ping to 50.0.0.5 it should reach Server0. Note that, in Router0, there should be static routing to 50.0.0.4/30 network.

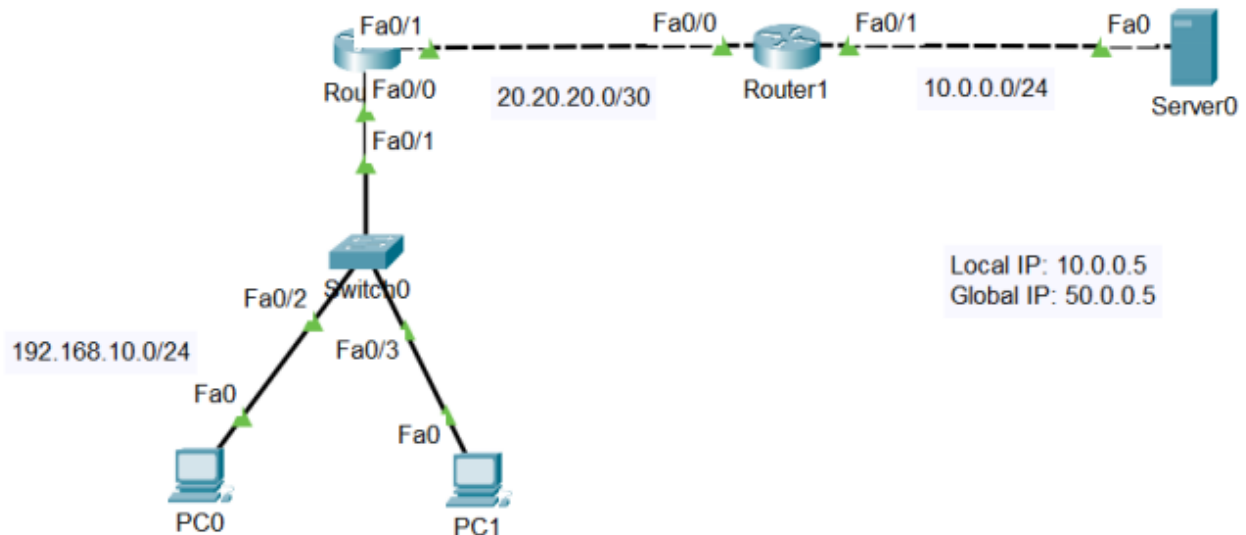


Figure 3: A sample network topology for the configuration of NAT.

1. Configure R0 Interfaces

```

1 Router(config)# int fa0/0
2 Router(config-if)# ip address 192.168.10.1 255.255.255.0
3 Router(config-if)# no shutdown
4 Router(config)# int fa0/1
5 Router(config-if)# ip address 20.20.20.1 255.255.255.252
6 Router(config-if)# no shutdown
7 Router(config-if)# exit
8 Router(config)# ip route 50.0.0.4 255.255.255.252 20.20.20.2
9 Router(config)# exit
10 Router# copy running-config startup-config

```

2. Configure R1 Interfaces

```

1 Router(config)# int fa0/1
2 Router(config-if)# ip address 10.0.0.1 255.255.255.0
3 Router(config-if)# no shutdown
4 Router(config)# int fa0/0
5 Router(config-if)# ip address 20.20.20.2 255.255.255.252
6 Router(config-if)# no shutdown
7 Router(config-if)# exit
8 Router(config)# ip route 192.168.10.0 255.255.255.0 20.20.20.1
9 Router(config)# exit
10 Router# copy running-config startup-config

```

3. Configure PC0

```

1 IP: 192.168.10.5
2 Mask: 255.255.255.0
3 Gateway: 192.168.10.1

```

4. Configure PC1

```

1 IP: 192.168.10.10
2 Mask: 255.255.255.0
3 Gateway: 192.168.10.1

```

5. Configure Server0

```

1 IP: 10.0.0.5
2 Mask: 255.255.255.0
3 Gateway: 10.0.0.1

```

6. Enable static NAT insider Router1

```

1 Router(config)# ip nat inside source static 10.0.0.5 50.0.0.5
2 Router(config)# int fa0/1
3 Router(config-if)# ip nat inside
4 Router(config)# int fa0/0
5 Router(config-if)# ip nat outside
6 Router(config-if)# end
7 Router# copy running-config startup-config

```

7. Verify NAT

```

1 Router# show ip nat translations
2 Router# show ip nat statistics

```

Part H: Configure Dynamic NAT

The exact network topology with the same device models, as shown in [Figure 3](#), has been used for this configuration.

Here the objective is the private address block 192.168.10.0 network will be converted to public address range 50.0.0.1-3/30. So, NAT will be applied on the Router 0.

The parts **Configure R1 Interfaces**, **Configure all the PCs and Server0**, and **Verify NAT** are the same as the previous section . The only difference is in the configuration of NAT inside **Router0**.

1. Enable dynamic NAT insider Router0

Create an ACL that permits all the IP addresses from the network 192.168.10.0/24.

```

1 Router(config)# access-list 1 permit 192.168.10.0 0.0.0.255

```

Create a NAT Pool for the Global IP Addresses from the network 50.0.0.0/30

```

1 Router(config)# ip nat pool NAT_Pool 50.0.0.1 50.0.0.3 netmask
  255.255.255.252

```

Associate the ACL to the NAT pool and configure the interfaces with appropriate **inside** and **outside** NAT commands.

```

1 Router(config)#ip nat inside source list 1 pool NAT_Pool
2 Router(config)# int fa0/1
3 Router(config-if)# ip nat inside

```

```
4 Router(config)# int fa0/0
5 Router(config-if)# ip nat outside
6 Router(config-if)# end
7 Router# copy running-config startup-config
```

Student ID:

Lab 05 Task

Your company, Make Abdullahpur Great Again (MAGA), is expanding its headquarters and requires a new network design to support various departments. The MAGA headquarters consists of the following departments across multiple floors: Engineering - Floor 1, Marketing - Floor 2, Sales - Floor 3, Administration - Floor 4, Management - Floor 5

Suppose they all share the prefix **192.(i).(i+3) % 10**. Below are the required devices for each of the departments:

$E(i) = (5i + 3) \bmod 41 + 5$	(Engineering)
$M(i) = (7i + 11) \bmod 26 + 5$	(Marketing)
$S(i) = (3i + 7) \bmod 36 + 5$	(Sales)
$A(i) = (11i + 17) \bmod 18 + 5$	(Administration)
$G(i) = (13i + 1) \bmod 9 + 5$	(Management)

where i is the last two digits your student ID

Implement and configure the topology, the devices should get their IP automatically from the DHCP server. Now for Engineering, Marketing and Sales the devices should use a separate DHCP server for automatic IP assignment & for Admin and Management they should use a router as a DHCP server. For each network you need to add minimum 4 end devices.

Use the same topology and configuration as the previous lab.

Now, your company is connected to the internet with another router called the Internet router. The number of global IP you have purchased is only 3, in which your own PC's local IP address has it's own global IP address, the rest of the devices uses the rest two IP addresses for accessing the internet. The global ip address starts from 110.i.i+1.0/24.

Lab 05 Grading Criteria

1. **Correct Topology & Router Configuration (15 pts)**
 - Correct topology (5 pts)
 - Correct router configuration (10 pts)
2. **NAT Configuration (20 pts)**
 - Correct Static NAT Configuration (5 pts)
 - Correct Non-Static NAT Configuration (10 pts)
3. **Routing (5 pts)**
4. **Total: 35 pts**